1 Camera Selection

Camera candidates (more can be added):

- https://www.3d-plus.com/products/caspex-4mpx-space-camera-heads/
- https://softwareservices.flir.com/BFS-U3-51S5/latest/Model/spec.html
- https://docs.baslerweb.com/daa2448-70um
- https://navi.ion.org/content/71/3/navi.661

Requirements:

- The camera must have a global shutter.
- The camera must have a resolution of at least 2000 pixels in the shortest side of the frame.
- The camera must focus a 10x10x15cm CubeSat and its background (infinite DOF).
- The sum of the volume of the camera and all its accessories must be less than a 10x10x5cm volume. The height of the camera plus any accesories stacked on it must be less than 5cm. The optical axis must be perpendicular to the 10x10cm base.
- The area of the smallest sphere that contains the CubeSat must occupy between a_{max} and 10% of the frame area, assuming that the sphere is centered on the frame. At least n occupancy percentages must be possible, where the first falls between 10 and $10 + (a_{max} - 10)/n\%$, the second between $10 + (a_{max} - 10)/n\%$ and $10 + 2 * (a_{max} - 10)/n\%$, and so forth. a_{max} is the smallest number between 30% and the largest occupancy allowed by the sphere dimensions and other constraints explicitly stated.

Calculations use $n = 4$, giving occupancy bands of 10–15, 15–20, 20–25, and 25–30%. The estimates allow up to four pixels of geometric defocus and require optical calibration.

Scenario A The camera focus is fixed, but the distance between the camera and the subject can be controlled externally. The distance range must be as narrow as possible, and the maximum distance as small as possible. No camera accesories can be added apart from the fixed lens.

- Basler dart daA2448-70uc, CS-mount version, global shutter, 2448×2048 native pixels. Use a fixed 2048×2048 native crop to fit the lens image circle.
- Raspberry Pi 6mm CS-mount wide-angle lens, with iris locked at a calibrated f/8 and focus fixed at approximately 0.60m.
- Nominal distance range 0.407–0.520m from the lens optical reference plane to the CubeSat centre. Distances of 520, 480, 440, and 407mm give occupancies of 15.0, 17.7, 21.3, and 25.1%, respectively.
- Nominal axial height is $20 + 34 - 5 = 49$mm, accounting for the lens's 5mm rear protrusion into the camera. A 49.6mm budget adds 0.5mm for mounting and 0.1mm for focus travel.
- Camera and lens volume is approximately $41\,\mathrm{cm}^3$, or $141\,\mathrm{cm}^3$ with a $100\,\mathrm{cm}^3$ integration allowance.

Table 1: Uncompressed image sizes for the output modes of the camera selected for Scenario A.

Output format	**Bits per pixel transmitted**	**Uncompressed** 2048×2048 **image**
BayerRG8	8	4 MiB
BayerRG12p	12	6 MiB
BayerRG12	16	8 MiB
RGB8 / BGR8	24 (8 per channel)	12 MiB
YCbCr422_8	16 on average	8 MiB

Scenario B The camera distance is fixed at 2m from the CubeSat. Any camera accesories can be added.

- FLIR BFS-U3-244S8M-C camera, global shutter, 5320×4600 pixels.
- Fujinon HF35XA-5M 35mm C-mount lens at f/8, focused at infinity.
- Optotune EL-16-40-GTC tunable lens, VIS 5D, Class 1, with M25.5 adapter, EL-E-4i controller, and six-pin Hirose cable.
- Top view packaging requires a folded optical path and remains unverified. The earlier 48mm height applied to a horizontal optical axis.
- Software for motion registration, focus fusion over approximately 1.9m to infinity, and centred cropping. Square crops of 3200, 2800, 2480, and 2240 pixels per side give occupancies of approximately 13.3, 17.4, 22.2, and 27.2%, respectively.

The estimated volume budget is $345\,\mathrm{cm}^3$, including $145\,\mathrm{cm}^3$ for the camera, optics, and lens controller, $100\,\mathrm{cm}^3$ for a dedicated computer (i.e. Raspberry Pi), and $100\,\mathrm{cm}^3$ for power, cabling, and mounting.